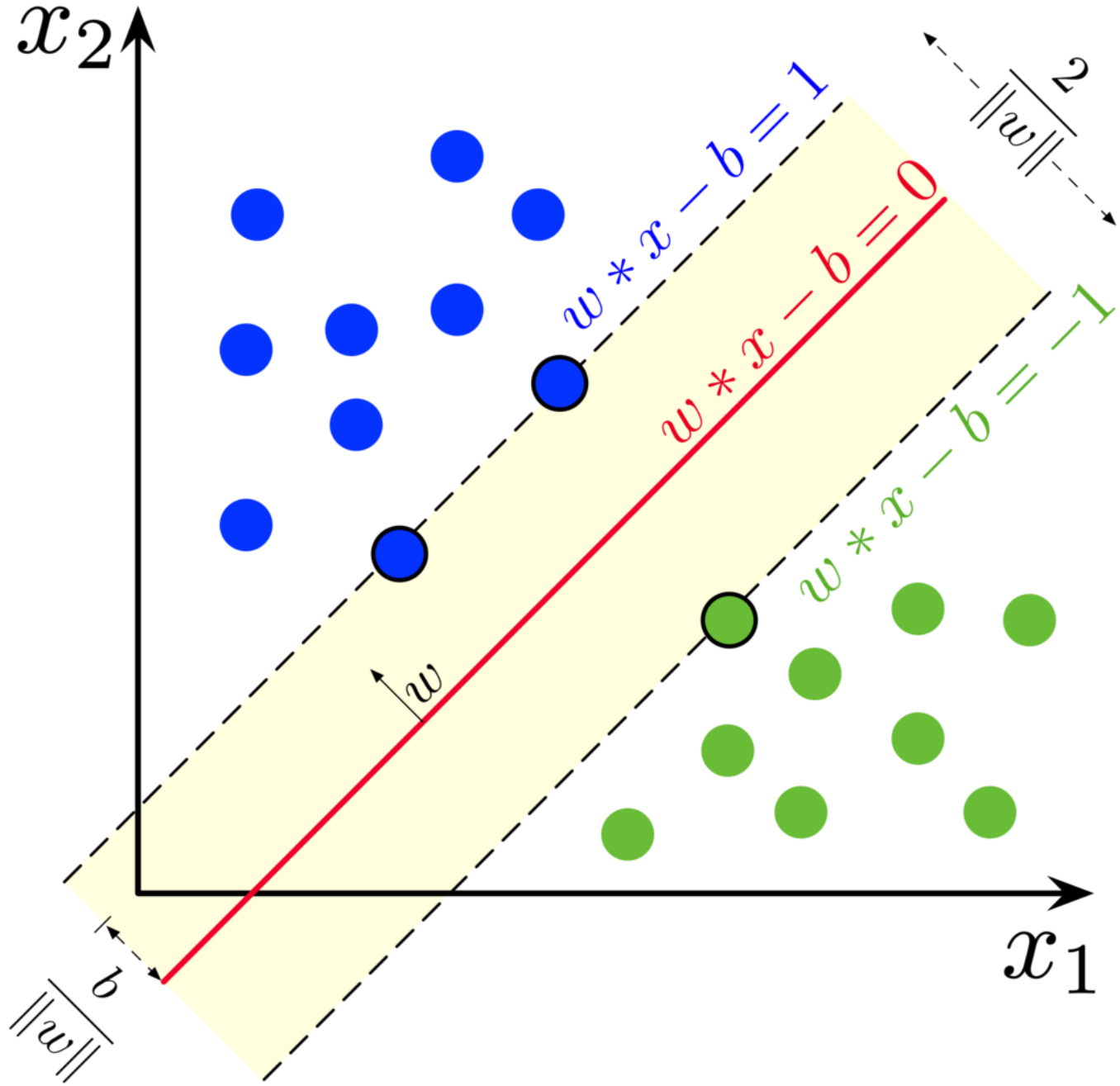
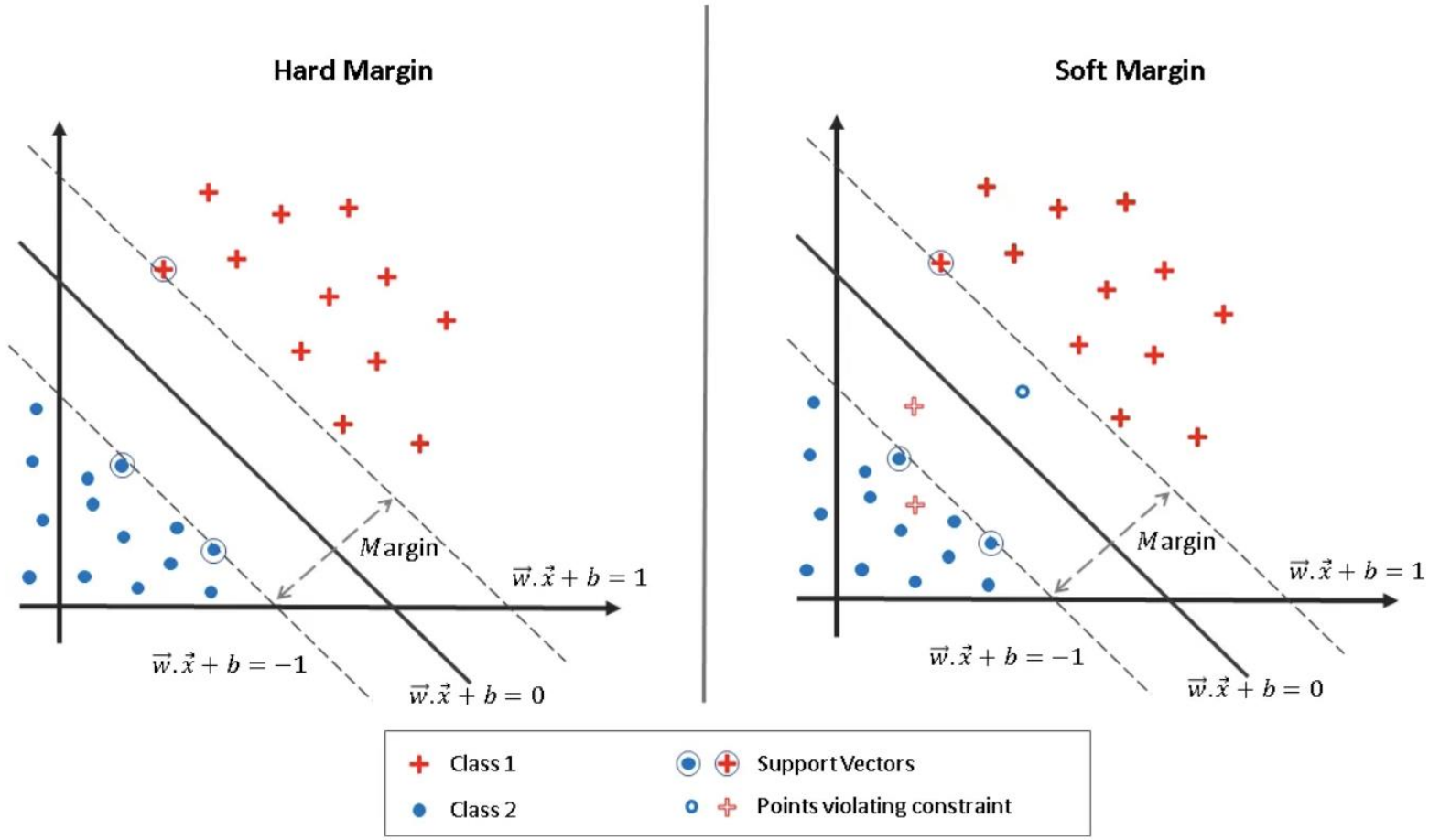
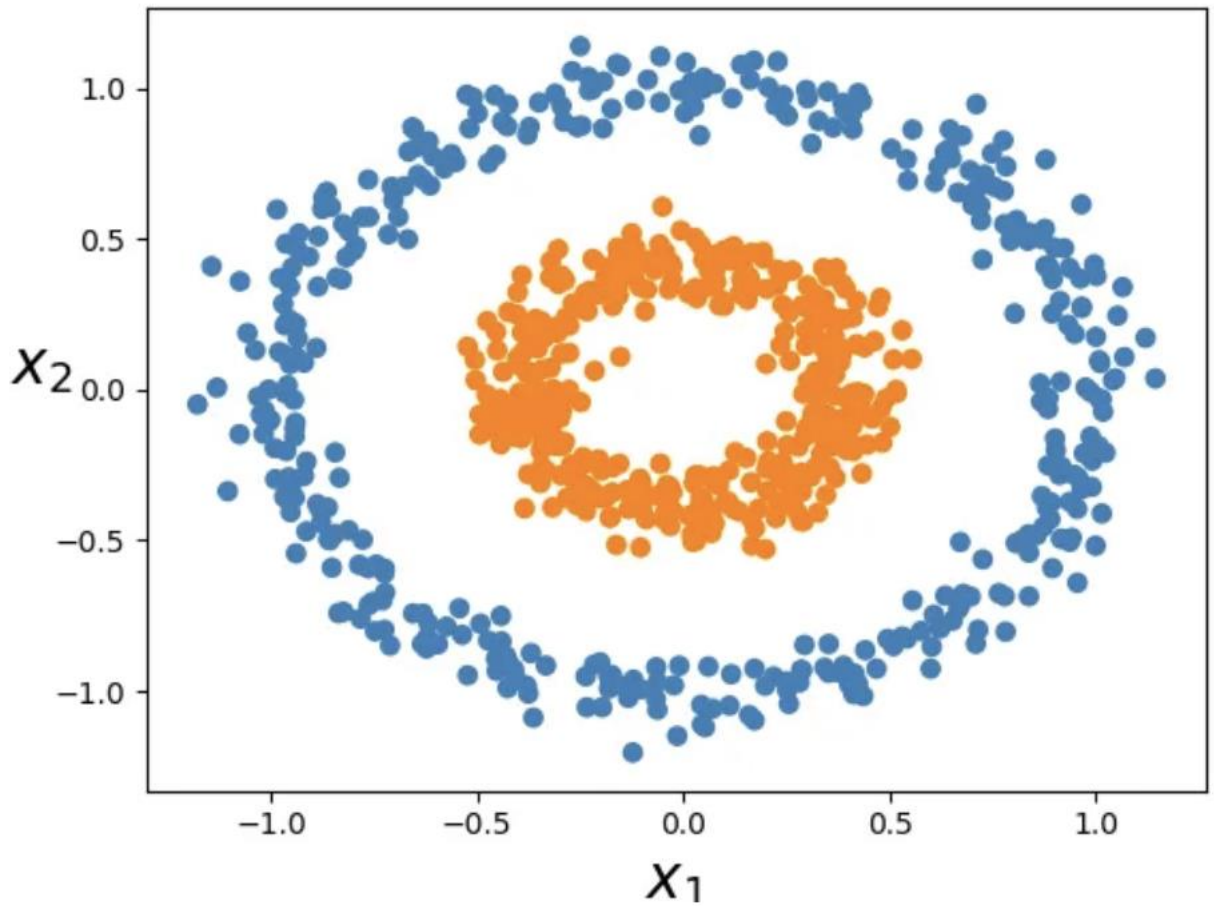
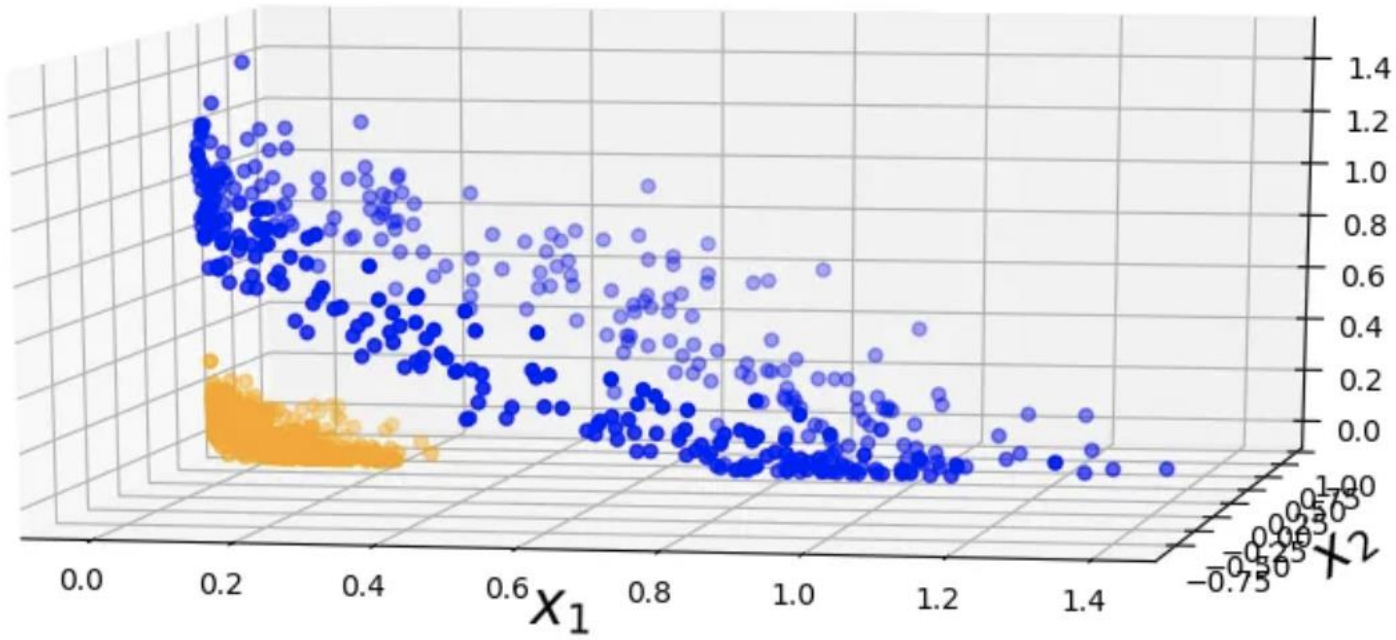
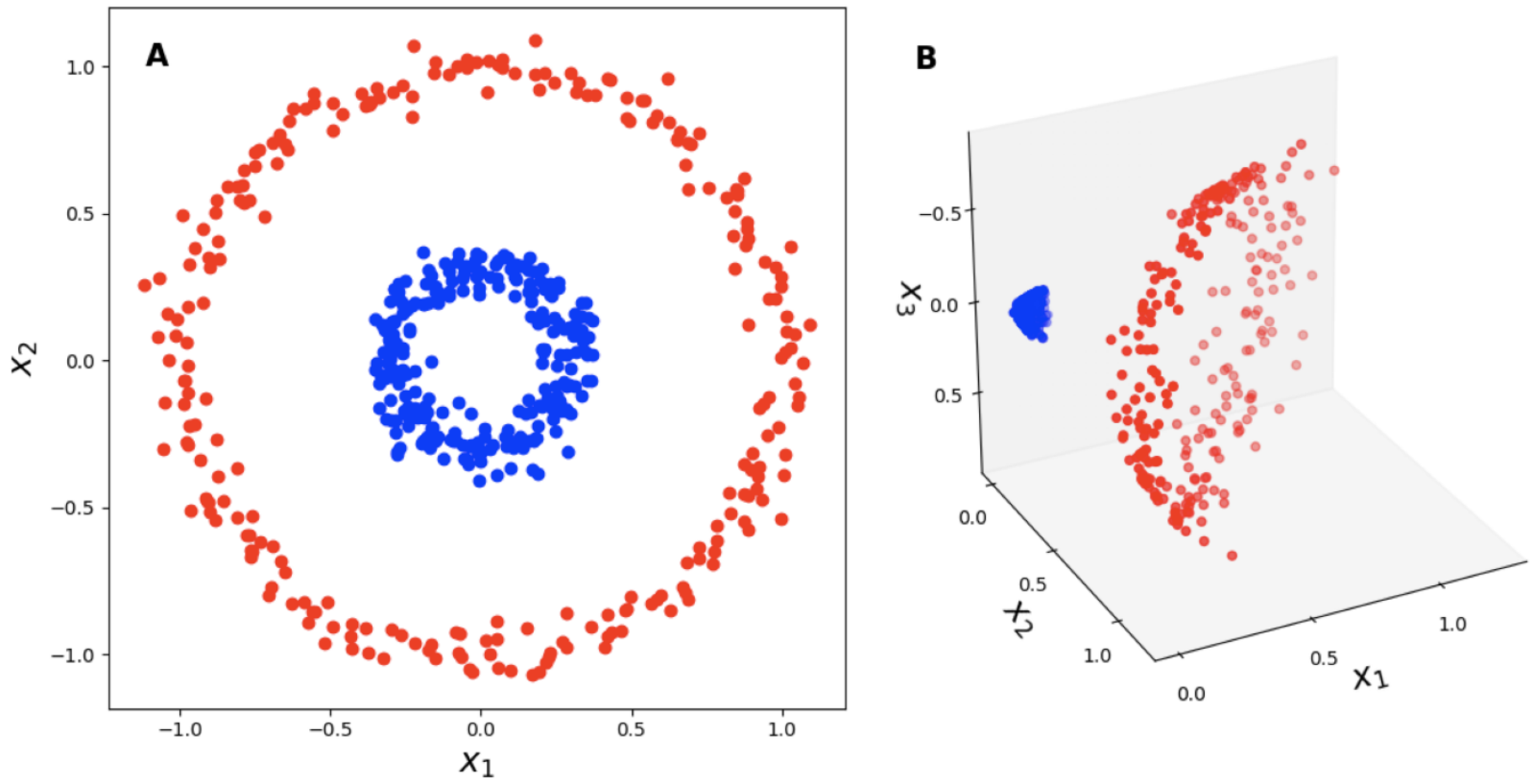

x_2
$w * x - b = 1$
$w * x - b = 0$
$w * x - b = -1$
$\frac{2}{\|w\|}$
w
$\frac{b}{\|w\|}$
x_1

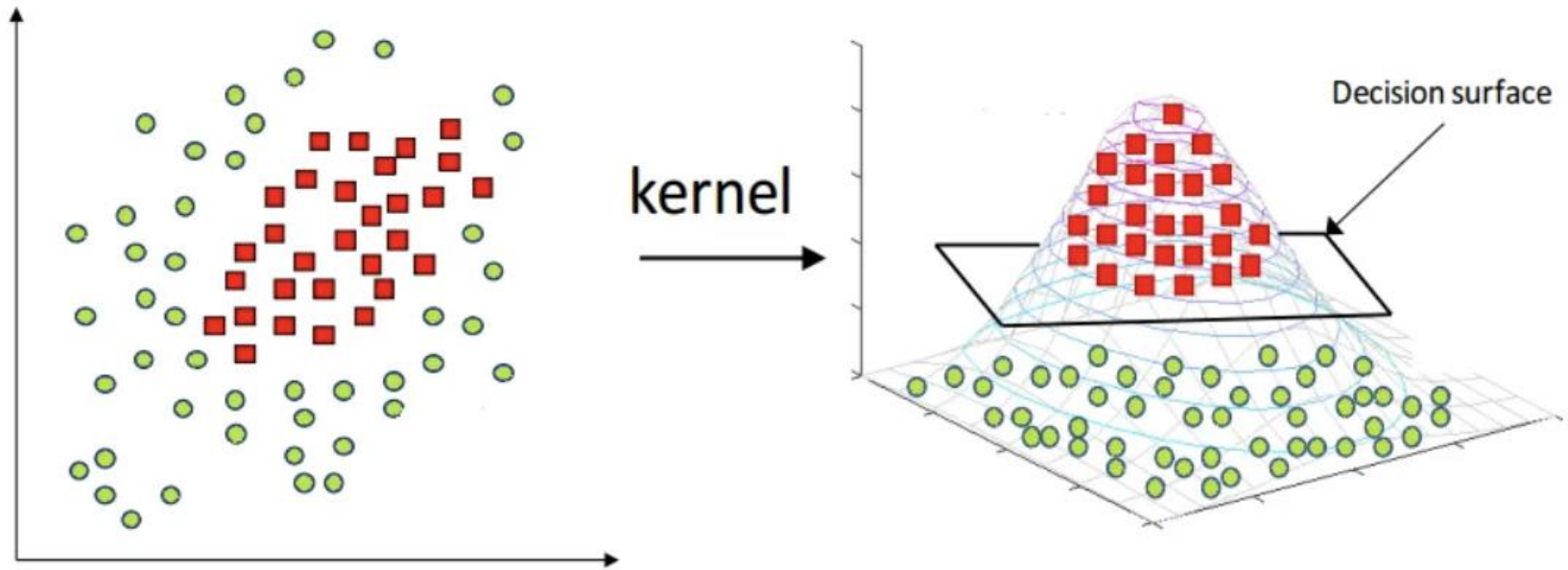

Hard Margin
Soft Margin
Margin
Margin
$\vec{w}.\vec{x}+b=1$
$\vec{w}.\vec{x}+b=-1$
$\vec{w}.\vec{x}+b=0$
$\vec{w}.\vec{x}+b=1$
$\vec{w}.\vec{x}+b=-1$
$\vec{w}.\vec{x}+b=0$
Class 1
Class 2
Support Vectors
Points violating constraint

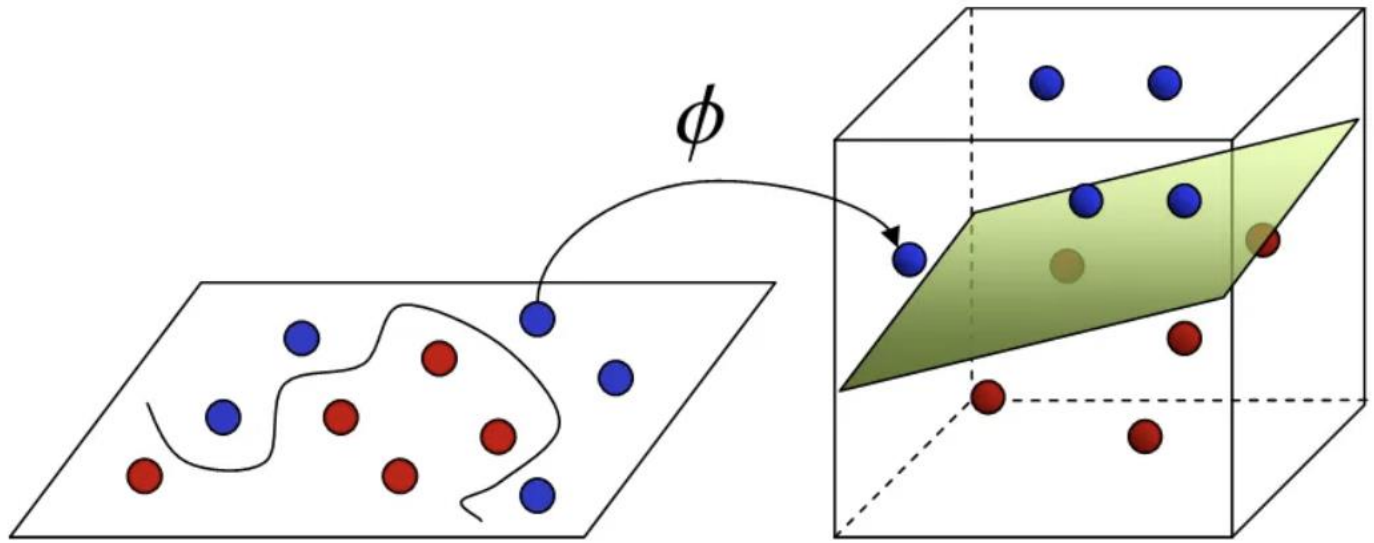

1.0
0.5
0.0
−0.5
−1.0
x_2
−1.0
−0.5
0.0
0.5
1.0
x_1

$$\phi(\mathbf{x}) = \phi\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} {x_1}^2 \\ \sqrt{2}\, x_1 x_2 \\ {x_2}^2 \end{pmatrix}$$


1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
x_1
0.8
1.0
1.2
1.4
−0.75
x_2


A
B
1.0
0.5
0.0
−0.5
−1.0
x_2
x_1
x_3
x_4


kernel
Decision surface


ϕ
Input Space
Feature Space